

Roots and Nature of Roots

1. If $\cos \theta$, $\sin \phi$ and $\sin \theta$ are in GP, then roots of $x^2 + 2\cot \phi \cdot x + 1 = 0$ are always

- (a) equal (b) real
(c) imaginary (d) greater than 1

2. If l, m and n are real, $l \neq m$, then the roots of the equation

$$(l - m)x^2 - 5(l + m)x - 2(l - m) = 0$$

- (a) real and equal (b) complex
(c) real and unequal (d) None of these

3. Let α and β be the roots of the equation $(x - a)(x - b) = c, c \neq 0$. Then, the roots of the equation $(x - \alpha)(x - \beta) + c = 0$ are

- (a) a, c (b) b, c
(c) a, b (d) $a + c, b + c$

4. If the roots of the equation $ax^2 + x + b = 0$ are real, then the roots of the equation $x^2 - 4\sqrt{ab}x + 1 = 0$ will be

- (a) rational (b) irrational
(c) real (d) imaginary

5. If $\frac{1}{3-4i}$ is a root of $ax^2 + bx + c = 0, (a, b, c \in \mathbf{R}, a \neq 0)$, then

- (a) $b + 6c = 0$ (b) $b = 6c$
(c) $a + 25c = 0$ (d) $b^2 = 6c$

6. If $1 - p$ is a root of the quadratic eq $px + 1 - p = 0$, then its roots are

- (a) 0, 1 (b) -1, 1
(c) 0, -1 (d) -2, 1

7. If a and b are rational and b is not a perfect square, then the quadratic equation with rational coefficients whose one root is $\frac{1}{a+\sqrt{b}}$, is

- (a) $x^2 - 2ax + (a^2 - b) = 0$
(b) $(a^2 - b)x^2 - 2ax + 1 = 0$
(c) $(a^2 - b)x^2 - 2bx + 1 = 0$
(d) None of the above

8. Let $p, q \in \mathbf{R}$. If $2 - \sqrt{3}$ is a root of the quadratic equation, $x^2 + px + q = 0$, then

- (a) $q^2 - 4p - 16 = 0$
(b) $p^2 - 4q - 12 = 0$
(c) $p^2 - 4q + 12 = 0$
(d) $q^2 + 4p + 14 = 0$

9. The number of integral values of m for which equation $(1 + m^2)x^2 - 2(1 + 3m)x + (1 + 8m) =$

0, has no real root is

- (a) 3
(b) infinitely many
(c) 1
(d) 2

10. If $a \neq b$, then the roots of the equation

$$2(a^2 + b^2)x^2 + 2(a + b)x + 1 = 0$$

- (a) real and distinct
(b) real and equal
(c) imaginary
(d) None of the above

11. How many roots of the equation $x - \frac{2}{x-1} = 1 - \frac{2}{x-1}$ have?

- (a) One
(b) Two
(c) Infinite
(d) None of these

12. The roots of the equation $x^4 - 8x^2 - 9 = 0$ are

- () i
() $\pm i$
() None of these

13. If the roots of the equation $8x^3 - 14x^2 + 7x - 1 = 0$ are in GP, then the roots are

- (a) $1, \frac{1}{2}, \frac{1}{4}$
(b) 2, 4, 8
(c) 3, 6, 12
(d) None of these

14. If $2 + i\sqrt{3}$ is a root of the equation $x^2 + px + q = 0$, where p and q are real, then (p, q) is equal to

- (a) (-4, 7)
(b) (4, -7)
(c) (4, 7)
(d) (-4, -7)

15. The number of real roots of $3^{2x^2-7x+7} = 9$ is

- (a) 0
(b) 2
(c) 1
(d) 4

16. If $2\sin^2 \frac{\pi}{8}$ is a root of the equation $x^2 + ax + b = 0$, where a and b are rational numbers, then $a - b$ is equal to

- (a) $-\frac{5}{2}$



- (b) $-\frac{3}{2}$
 (c) $-\frac{1}{2}$
 (d) $\frac{1}{2}$

17. The number of all possible positive integral values of α for which the roots of the quadratic equation, $6x^2 - 11x + \alpha = 0$ are rational numbers is

- (a) 5
 (b) 2
 (c) 4
 (d) 3

18. If the roots of $(a^2 + b^2)x^2 - 2(bc + ad)x + c^2 + d^2 = 0$ are equal, then

- (a) $\frac{a}{b} = \frac{c}{d}$
 (b) $\frac{a}{c} + \frac{b}{d} = 0$
 (c) $\frac{a}{d} = \frac{b}{c}$
 (d) $a + b = c + d$

19. Let $f(x) = x^2 + ax + b$; $a, b \in \mathbb{R}$. If $f(1) + f(2) + f(3) = 0$, then the roots of the equation $f(x) = 0$

- (a) are imaginary
 (b) are real and equal
 (c) are from the set $\{1, 2, 3\}$
 (d) real and distinct

20. If $\sin \alpha, \sin \beta$ and $\cos \alpha$ are in GP of $x^2 + 2x \cot \beta + 1 = 0$ are always

- (a) real
 (b) real and negative
 (c) greater than one
 (d) non-real

21. The solution set of the equation $pqx^2 - (p + q)^2x + (p + q)^2 = 0$ is

- (a) $\left\{\frac{p}{q}, \frac{q}{p}\right\}$
 (b) $\left\{pq, \frac{p}{q}\right\}$
 (c) $\left\{\frac{q}{p}, pq\right\}$
 (d) $\left\{\frac{p+q}{p}, \frac{p+q}{q}\right\}$

22. If the roots of the equation $qx^2 + px + q = 0$ are complex, where p and q are real, then the roots of the equation $x^2 - 4qx + p^2 = 0$ are

- (a) real and unequal
 (b) real and equal

- (c) imaginary
 (d) None of these

23. If $a < b < c < d$, then the roots of the equation $(x - a)(x - c) + 2(x - b)(x - d) = 0$ are

- (a) real and distinct
 (b) real and equal
 (c) imaginary
 (d) None of these

24. Two non-integer roots of the equation

$$(x^2 + 3x)^2 - (x^2 + 3x) - 6 = 0$$

are

- (a) $\frac{1}{2}(-3 + \sqrt{11}), \frac{1}{2}(-3 - \sqrt{11})$
 (b) $\frac{1}{2}(-3 + \sqrt{7}), \frac{1}{2}(-3 - \sqrt{7})$
 (c) $\frac{1}{2}(-3 + \sqrt{21}), \frac{1}{2}(-3 - \sqrt{21})$
 (d) none of these

25. The number of real values of x satisfying the eq

$$2\left(x^2 + \frac{1}{x^2}\right) - 9\left(x + \frac{1}{x}\right) + 14 = 0$$

- (a) 1
 (b) 2
 (c) 3
 (d) 4

26. The number of solutions of $\sqrt{4 - x} + \sqrt{x + 9} = 5$ is:

- (a) 0
 (b) 1
 (c) 2
 (d) 3

27. The number of solutions of

$$\sqrt{3x^2 + x + 5} = x - 3, \text{ where } x \in \mathbb{R}$$

is:

- (a) 0
 (b) 1
 (c) 2
 (d) 4

28. The number of solutions of $\sqrt{x + 1} - \sqrt{x - 1} = 1$ ($x \in \mathbb{R}$) is

- (a) 1
 (b) 2
 (c) 4
 (d) infinite



Roots Under Particular Conditions

29. If the roots of the equation $\frac{\alpha}{x-\alpha} + \frac{\beta}{x-\beta} = 1$ be equal in magnitude but opposite in sign, then $\alpha + \beta$ is equal to

- (a) 0
- (b) 1
- (c) 2
- (d) None of these

30. If the roots of the equation $\frac{1}{x+p} + \frac{1}{x+q} = \frac{1}{r}$ are equal in magnitude but opposite in sign, then the product of the roots will be

- (a) $\frac{p^2+q^2}{2}$
- (b) $-\frac{(p^2+q^2)}{2}$
- (c) $\frac{p^2-q^2}{2}$
- (d) $-\frac{(p^2-q^2)}{2}$

31. The roots of the given equation $(p-q)x^2 + (q-r)x + (r-p) = 0$ are

- (a) $\frac{p-q}{r-p}, 1$
- (b) $\frac{q-r}{p-q}, 1$
- (c) $\frac{r-p}{p-q}, 1$
- (d) $1, \frac{q-r}{p-q}$

32. If $a > 0, b > 0, c > 0$, then both the roots of the equation $ax^2 + bx + c = 0$

- (a) are real and negative
- (b) have negative real part
- (c) are rational numbers
- (d) None of these

33. If roots of the equation $(a-b)x^2 + (c-a)x + (b-c) = 0$ are equal, then a, b and c are in

- (a) AP
- (b) HP
- (c) GP
- (d) None of these

34. If the roots of the equation $(p^2 + q^2)x^2 - 2q(p+r)x + (q^2 + r^2) = 0$ be real and equal, then p, q and r will be in

- (a) AP
- (b) GP

- (c) HP
- (d) None of these

35. Suppose $p \in \mathbf{R}$. If the roots of $3x^2 + 2x + p(p-1) = 0$ are of opposite signs, then p must lie in the interval

- (a) $(-\infty, 0)$
- (b) $(0, 1)$
- (c) $(1, \infty)$
- (d) $(2, \infty)$

36. Suppose $a \in \mathbf{R}$. If $3x^2 + 2(a^2 + 1)x + (a^2 - 3a + 2) = 0$ possesses roots of opposite signs, then a lies in the interval:

- (a) $(-\infty, -1)$
- (b) $(-1, 1)$
- (c) $(1, 2)$
- (d) $(2, 3)$

37. The number of values of a for which $(a^2 - 3a + 2)x^2 + (a^2 - 5a + 6)x + a^2 - 4 = 0$ is an integer in x , is

(
(

3

Relations Between Roots and Coefficients & Formation of Q.E.

38. The quadratic equation whose roots are the AM and HM of the roots of the equation $x^2 + 7x - 1 = 0$ is

- (a) $14x^2 + 14x - 45 = 0$
- (b) $14x^2 - 14x + 14 = 0$
- (c) $14x^2 + 45x - 14 = 0$
- (d) None of these

39. If α and β are the roots of $ax^2 + 2bx + c = 0$, then $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$ is equal to

- (a) $\frac{4b^2-2ac}{ac}$
- (b) $\frac{4b^2-4ac}{ac}$
- (c) $\frac{2b^2-2ac}{ac}$
- (d) $\frac{2b^2-4ac}{ac}$

40. If α and β are the roots of the equation $px^2 + qx + r = 0$, then the value of $\alpha^3\beta + \beta^3\alpha$ is

- (a) $\frac{r}{p^3}$



(b) $\frac{r(q^2-2pr)}{p^3}$

(c) $\frac{r(q^2+2pr)}{p^3}$

(d) $\frac{(q^2-2pr)}{p^3}$

41. If α, β and γ are the roots of the equation $x(1+x^2) + x^2(6+x) + 2 = 0$, then the value of $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$ is

(a) -3

(b) $\frac{1}{2}$

(c) $-\frac{1}{2}$

(d) None of these

42. If the roots of the equation $x^2 + px + q = 0$ are in the same ratio as those of the equation $x^2 + lx + m = 0$, then p^2m is equal to

(a) lq

(b) l^2q

(c) lq^2

(d) None of the above

43. If α and β are the roots of equation $2x^2 - 5x + 7 = 0$, then the equation whose roots are $2\alpha + 3\beta$ and $3\alpha + 2\beta$, is

(a) $2x^2 - 25x + 82 = 0$

(b) $2x^2 + 25x + 82 = 0$

(c) $x^2 - 25x + 82 = 0$

(d) None of these

44. If α and β are the roots of the equation $7x^2 - 3x - 2 = 0$, then the value of $\frac{\alpha}{1-\alpha^2} + \frac{\beta}{1-\beta^2}$ is equal to

(a) $\frac{27}{32}$

(b) $\frac{1}{24}$

(c) $\frac{3}{8}$

(d) $\frac{27}{16}$

45. If m is chosen in the quadratic equation $(m^2 + 1)x^2 - 3x + (m^2 + 1)^2 = 0$ such that the sum of its roots is greatest, then the absolute difference of the cubes of its roots is

(a) $10\sqrt{5}$

(b) $8\sqrt{5}$

(c) $8\sqrt{3}$

(d) $4\sqrt{3}$

46. If the roots of $a_1x^2 + b_1x + c_1 = 0$ are α_1, β_1 and that of $a_2x^2 + b_2x + c_2 = 0$ are α_2, β_2 such that $\alpha_1\alpha_2 = \beta_1\beta_2 = 1$, then

(a) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

(b) $\frac{a_1}{c_2} = \frac{b_1}{b_2} = \frac{c_1}{a_2}$

(c) $a_1a_2 = b_1b_2 = c_1c_2$

(d) None of the above

47. If α, β are the roots of $ax^2 + bx + c = 0 (a \neq 0)$ and $\alpha + h, \beta + h$ are the roots of $px^2 + qx + r = 0 (p \neq 0)$, then the ratio of the squares of their discriminants is

(a) $a^2 : p^2$

(b) $a : p^2$

(c) $a^2 : p$

(d) $a : 2p$

48. $\tan \alpha$ and $\tan \beta$ are the roots of the equation $x^2 + ax + b = 0$, then the value of $\sin^2(\alpha + \beta) + \cos^2(\alpha + \beta) + b \cos^2(\alpha + \beta)$ is equal to

(a) $\frac{b}{a}$

49. The coefficient of x in the equation $x^2 + px + q = 0$ was taken as 17 in place of 13 its roots were found to be -2 and -15. The roots of the original equation are

(a) 3, 10

(b) -3, -10

(c) -5, -8

(d) None of these

50. Two students while solving a quadratic equation in x , one copied the constant term incorrectly and got the roots 3 and 2. The other copied the constant term coefficient of x^2 correctly as -6 and 1 respectively the correct roots are

(a) 3, -2

(b) -3, 2

(c) -6, -1

(d) 6, -1

51. Let α and β be the roots of equation $px^2 + qx + r = 0$ $p \neq 0$. If p, q and r are in AP and $\frac{1}{\alpha} + \frac{1}{\beta} = 4$, then the value of $|\alpha - \beta|$ is

(a) $\frac{\sqrt{61}}{9}$



(b) $\frac{2\sqrt{17}}{9}$

(c) $\frac{\sqrt{34}}{9}$

(d) $\frac{2\sqrt{13}}{9}$

52. If the equation $x^4 - 4x^3 + ax^2 + bx + 1 = 0$ has four positive roots, then a, b are

(a) $a = 4, b = 6$

(b) $a = -4, b = 6$

(c) $a = 2, b = 3$

(d) $a = 6, b = -4$

53. If $\sin \alpha$ and $\cos \alpha$ are the roots of the equation $ax^2 + bx + c = 0$, then

(a) $a^2 - b^2 + 2ac = 0$

(b) $(a - c)^2 = b^2 + c^2$

(c) $a^2 + b^2 - 2ac = 0$

(d) $a^2 + b^2 + 2ac = 0$

54. If α and β be the roots of the equation $2x^2 + 2(a + b)x + a^2 + b^2 = 0$, then the equation whose roots are $(\alpha + \beta)^2$ and $(\alpha - \beta)^2$, is

(a) $x^2 - 2abx - (a^2 - b^2)^2 = 0$

(b) $x^2 - 4abx - (a^2 - b^2)^2 = 0$

(c) $x^2 - 4abx + (a^2 - b^2)^2 = 0$

(d) None of the above

55. Let α and α^2 be the roots of $x^2 + x + 1 = 0$ then the equation whose roots are α^3

(a) $x^2 - x + 1 = 0$

(b) $x^2 + x - 1 = 0$

(c) $x^2 + x + 1 = 0$

(d) $x^{60} + x^{30} + 1 = 0$

56. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then the equation whose roots are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$, is

(a) $acx^2 + (a + c)bx + (a + c)^2 = 0$

(b) $abx^2 + (a + c)bx + (a + c)^2 = 0$

(c) $acx^2 + (a + b)cx + (a + c)^2 = 0$

(d) None of the above

57. The equation whose roots are the square of the roots of the equation $2x^2 + 3x + 1 = 0$, is

(a) $4x^2 + 5x + 1 = 0$

(b) $4x^2 - x + 1 = 0$

(c) $4x^2 - 5x - 1 = 0$

(d) $4x^2 - 5x + 1 = 0$

58. If α and β be two roots of the equation $x^2 - 64x + 256 = 0$. Then, value of $\left(\frac{\alpha^3}{\beta^5}\right)^{1/8} + \left(\frac{\beta^3}{\alpha^5}\right)^{1/8}$ is

(a) 2

(b) 3

(c) 1

(d) 4

59. If $\tan 25^\circ$ and $\tan 20^\circ$ are roots of the quadratic equation $x^2 + 2px + q = 0$, then $2p - q$ is equal to

(a) -2

(b) -1

(c) 0

(d) 1

60. Suppose $0 < \alpha < \beta$, and $2\alpha + \beta = \pi/2$. If $\tan \alpha, \tan \beta$ are roots of $ax^2 + bx + c = 0$, then

(a) $\tan \alpha = \frac{c-a}{b}$

(b) $\tan \beta = \frac{c+a}{b}$

(c) $\frac{b}{c-a}$

(d) $= -\frac{b}{c+a}$

Condition for Common Roots

61. Let $\lambda \neq 0$ be in R . If α and β are the roots of the equation, $x^2 - x + 2\lambda = 0$ and α and γ are the roots of the equation, $3x^2 - 10x + 27\lambda = 0$, then

$\frac{\beta\gamma}{\lambda}$ is equal to

(a) 36

(b) 9

(c) 27

(d) 18

62. If the equations $x^2 + 2x + 3 = 0$ and $ax^2 + bx + c = 0$; $a, b, c \in R$, have a common root, then $a : b : c$ is equal to

(a) 1 : 2 : 3

(b) 3 : 2 : 1

(c) 1 : 3 : 2

(d) 3 : 1 : 2

63. If the equation $2x^2 + 3x + 5\lambda = 0$ and $x^2 + 2x + 3\lambda = 0$ have a common root, then λ is equal to

(a) 0

(b) -1

(c) 0, -1

(d) 2, -1



64. If atleast one root of $2x^2 + 3x + 5 = 0$ and $ax^2 + bx + c = 0, a, b, c \in N$ is common, then the maximum value of $a + b + c$ is

- (a) 10
- (b) 0
- (c) does not exist
- (d) None of these

65. If the equations $2ax^2 - 3bx + 4c = 0$ and $3x^2 - 4x + 5 = 0$ have a common root, then $(a + b)/(b + c)$ is equal to $(a, b, c \in R)$

- (a) $1/2$
- (b) $3/35$
- (c) $34/31$
- (d) $29/31$

66. A value of b for which the equations $x^2 + bx - 1 = 0, x^2 + x + b = 0$ have one root in common, is

- (a) $-\sqrt{2}$
- (b) $-i\sqrt{3}$
- (c) $i\sqrt{5}$
- (d) $\sqrt{2}$

Sign of Quadratic Expression

67. The number of integral values of m for which the quadratic expression, $(1 + 2m)x^2 - 2(1 + x + 4(1 + m)), x \in R$, is always positive, is

- (a) 6
- (b) 8
- (c) 7
- (d) 3

68. The integer ' k ', for which the inequality $x^2 - 2(3k - 1)x + 8k^2 - 7 > 0$ is valid for every x in R is

- (a) 3
- (b) 2
- (c) 4
- (d) 0

69. The values of ' a ' for which $(a^2 - 1)x^2 + 2(a - 1)x + 2$ is positive for any x , are

- (a) $a \geq 1$
- (b) $a \leq 1$
- (c) $a > -3$
- (d) $a < -3$ or $a > 1$

Equations Containing Absolute Values.

70. The number of real solutions of the equation $|x|^2 - 3|x| + 2 = 0$ is

- (a) 4
- (b) 1
- (c) 3
- (d) 2

71. The product of the roots of the equation $9x^2 - 18|x| + 5 = 0$, is

- (a) $\frac{5}{27}$
- (b) $\frac{25}{9}$
- (c) $\frac{5}{9}$
- (d) $\frac{25}{81}$

72. The roots of the equation $|2x - 1|^2 - 3|2x - 1| + 2 = 0$ are

- (a) $\left\{-\frac{1}{2}, 0, \frac{1}{2}\right\}$
- (b) $\left\{-\frac{1}{2}, 0, \frac{3}{2}\right\}$
- (c) $\left\{-\frac{3}{2}, \frac{1}{2}, 0, 1\right\}$
- (d) $\left\{-\frac{1}{2}, 0, 1, \frac{3}{2}\right\}$

73. The number of real solutions of $x^2 + 5|x| + 4 = 0$ is

- (a) 0
- (b) 1
- (c) 2
- (d) 4

74. Sum of the roots of the equation $x^2 + |2x - 3| - 4$ is

- (a) 1
- (b) -2
- (c) 2
- (d) $-\sqrt{2}$

s Lies in the Interval

75. If both the roots of the quadratic equation $x^2 - mx + 4 = 0$ are real and distinct and they lie in the interval $[1, 5]$ then m lies in the interval

- (a) (4, 5)
- (b) $(-5, -4)$
- (c) (5, 6)
- (d) (3, 4)

